Ahmad Mujtaba

Generative AI Engineer

+91-98070-52640 | ahmad.iiitk@gmail.com | LinkedIn | GitHub | Portfolio

Professional Summary

Generative AI Engineer with 3+ years building LLM, agentic, and document intelligence solutions for healthcare and automotive teams, including health insurance and prior authorization workflows at Deloitte. Skilled in Python, RAG, agent frameworks (LangChain/LangGraph, MCP), and deploying AI-driven solutions on Azure, AWS, and Databricks within Agile environments. Recent results: policy entity extraction improved to 99% on internal evaluations, SOP-grounded agent task completion lifted from 38% to ∼80%, and prompt-token usage reduced by ∼40% in browser automation.

Technical Skills

Generative AI & LLMs: LLM applications, AI agents, RAG, Intelligent Document Processing, Prompt engineering, Model evaluation, Model Context Protocol (MCP), OpenAI, Anthropic Claude, Azure OpenAI, Azure Content Understanding, Google Gemini, LangChain, LangGraph, Milvus, FastAPI, Playwright

Machine Learning & Data: Python, SQL, PySpark, Databricks, Scikit-learn, TensorFlow, Predictive modeling, NLP, PaddleOCR, Layout analysis, Feature engineering, Data analysis

Data Engineering: ETL/ELT pipelines, Azure Databricks, Data modeling, Medallion architecture, MS SQL Server, Amazon S3, Schema mapping, Data quality, Workflow orchestration, API development

Platforms & Delivery: Azure App Services, Azure Cognitive Services, Azure Key Vault, AWS Lambda, CloudWatch, SageMaker, Docker, Git, GitLab, CI/CD, MLOps, Testing/QA, Agile, Power BI

Domain Experience: Healthcare, Prior Authorization, Health Insurance, Automotive, Manufacturing, FMCG, Warranty analytics, Conversational AI, Cross-functional delivery

Experience

Deloitte US-India — Gurugram, India
Generative AI Engineer — *July 2025 – Present*

- Designed and delivered an intelligent document processing pipeline for unstructured health insurance policy documents supporting prior authorization workflows, using PaddleOCR, layout-aware parsing, and LLM-based reasoning over policy clauses.
- Improved health policy entity extraction accuracy from ∼90% with a GPT-5.1 baseline to 99% on internal evaluations by moving to Gemini-3-Pro and adding a canonical-comparison validation loop.
- Contributed to a client prior authorization automation solution on Azure Databricks, performing AI-driven extraction from prior authorization documents using Azure Content Understanding with GPT-4.1; led the migration of the underlying model from GPT-4.1 to GPT-5.2, improving extraction accuracy and reducing hallucinations.
- Delivered an OpenAI Computer-Using Agent for a healthcare client to automate manual web-based workflows; deployed in a Docker environment on AWS with a streamed human-in-the-loop interface for reviewer oversight, execution monitoring, and testing/QA.
- Raised task completion on SOP-driven benchmark workflows from 38% to ∼80% by adding a Milvus-backed RAG layer that grounded responses in retrieved standard operating procedures.
- Built a Playwright-based MCP automation tool that used accessibility tree snapshots instead of raw DOM context, reducing prompt-token usage by ∼40% in internal testing.
- Built Databricks ETL pipelines in PySpark and Python for schema mapping, key reconciliation, and multi-table joins that supported downstream healthcare analytics and reporting.

Cognizant Technology Solutions — Noida, India
Associate Data Scientist — *Sep 2022 – May 2025*

- Developed warranty-claim classification models on Azure Databricks for an automotive client, improving recall and reducing manual review effort.

- Built a B2B conversational assistant using AWS Lex and Azure OpenAI to parse unstructured product orders and trigger reordering workflows through AWS Lambda.
- Improved reliability of Python services on Azure App Services by resolving timeout bottlenecks, supporting API development needs, and shifting backend tasks to AWS Lambda and S3.
- Created Power BI dashboards to track model performance and business KPIs for cross-functional stakeholders.

AiEnsured Remote, India
Machine Learning Engineer Intern *Jul 2021 – Aug 2021*

- Supported object-detection model development using CNN-based approaches, including code optimization and error analysis.
- Implemented regression and classification models on varied datasets and contributed to feature-engineering experiments that improved baseline performance.

Projects

CUA — Computer Using Agent Open Source
github.com/pypi-ahmad/computer-use *React, FastAPI, Docker, Playwright MCP, OpenAI, Claude, Gemini*

- Built a multi-provider Computer Use agent workbench supporting OpenAI GPT-5.5/5.4, Anthropic Claude Opus 4.7 / Sonnet 4.6, and Google Gemini 3 Flash from a single React/FastAPI UI, with all desktop actions executed inside an isolated Docker Ubuntu/XFCE sandbox.
- Implemented a two-phase provider-native web-search planning pass, stuck-agent detection via action fingerprinting, async operator safety-confirmation flow, and real-time WebSocket screenshot/log streaming.
- Hardened the API surface with loopback-only binding, host allowlist, CSRF/origin gate, per-IP rate limiting, and an in-container action allowlist; covered by 482 passing pytest tests across engine clients, schema validation, and action dispatch.

Education

Indian Institute of Information Technology (IIIT) Kurnool, India
M.Tech in Data Analytics and Decision Sciences *Oct 2020 – Jun 2022*

- Coursework: Machine Learning, Deep Learning, NLP, Computer Vision, Statistics.

Maulana Azad National Urdu University Hyderabad, India
B.Tech in Computer Science Engineering *Aug 2015 – Jun 2019*

- Coursework: Data Structures & Algorithms, Engineering Mathematics, Web Development.

Certifications

Machine Learning Specialization | *DeepLearning.AI via Coursera* Jan 2023